

Wenshuo Guo

CONTACT email: wguo@cs.berkeley.edu
<https://people.eecs.berkeley.edu/~wguo/>
mobile: +1-510-345-7311

EDUCATION **Ph.D. Candidate in Computer Science**
University of California, Berkeley
Fall 2018 - present

Advisor: [Michael I. Jordan](#)

Area: Machine Learning and Economics, Optimization

Affiliations: Berkeley AI Research (BAIR) Lab, Berkeley RISE lab, Berkeley CLIMB Center

Highlight courses:

Theoretical statistics (I, II), probability theory,
convex optimization (II, III), causal inference,
robust statistics, machine learning in social dynamics

B.S. in Physics, Computer Science and Mathematics (First class honors)
Hong Kong University of Science and Technology
2014 - 2018

Highlight graduate-level courses:

Computer vision, statistical mechanics, combinatorial optimization,
data mining, statistical learning for text data analytics

AWARDS	Google PhD Fellowship	2021–
	Facebook PhD Fellowship (<i>declined</i>)	2021
	Microsoft PhD Fellowship Finalist	2020
	Apple PhD Fellowship Finalist	2020
	UC Berkeley Gilbert Henry Gates Graduate Fellowship	2018-19
	HKUST Undergraduate Research Award - Champion of the Year	2017
	Hong Kong Government Scholarship Reaching Out Award	2017
	HKUST Physics Overseas Conference Travel Grant	2017
	HKUST Paul and May Chu Physics Research Award	2016
	HKUST Undergraduate Summer Research Fellowship	2015
	HKUST Physics Department Entry Scholarship	2015
	HKUST University Scholarship	2014-18

WORK
EXPERIENCE

	<i>Machine learning intern, CNN</i>	2022/01-2022/05
	• Worked on machine learning algorithms for large-scale news aggregation and personalized recommendation, hosted by Brian Whitman	
	<i>Research intern, Microsoft Research (New England)</i>	2021/05-2021/08
	• Machine learning and statistics team, hosted by David Alvarez-Melis • Worked on robust machine learning under domain shift in data • Designed light-weight algorithms to achieve domain adaptation without model re-training based on recent advances in optimal transport theory	
	<i>Undergraduate research assistant, HKUST</i>	2015-2018
	• Worked on graduate-level research projects in statistical physics, machine learning with educational data, and graph theory • Resulted in three publications at international conference venues • Hosted by Szeto Kwok Yip, Dit-Yan Yeung, and Mordecai J. Golin	
SERVICE	Program committee of International Conference on Algorithmic Learning Theory (ALT)	2022
	Co-President at UC Berkeley Women in Computer Science and Engineering (WICSE)	2021-22
	Program chair of WiML workshop at International Conference on Machine Learning (ICML)	2021
	Social chair at UC Berkeley Women in Computer Science and Engineering (WICSE)	2020-21
	Reviewer for Transactions in Machine Learning Research (TMLR)	2020-now
	Reviewer for International Conference on Artificial Intelligence and Statistics (AISTATS)	2021-
	Reviewer for International Conference on Machine Learning (ICML)	2020-
	Reviewer for Conference on (NeurIPS)	2020-now
	Reviewer for International Conference on Learning Representations (ICLR)	2020-now
	Reviewer for Conference on Computer Vision and Pattern Recognition (CVPR)	2020-now
	Reviewer for International Conference on Machine Learning (ICML)	2019-now
	Mentor at UC Berkeley CalMentor Program	2020-21
	Mentor at UC Berkeley AI Research mentor program	2019–

TEACHING

Data, Inference, and Decisions

Jacob Steinhardt, Moritz Hardt, 2020 Spring. (Undergraduate, upper division)

Statistical Learning Theory

Benjamin Recht, Moritz Hardt, 2019 Fall. (Graduate)

PUBLICATIONS (asterisk indicates joint or alphabetical authorship.)

1. **No-Regret Learning in Partially-Informed Auctions**
Wenshuo Guo, Michael I. Jordan, Ellen Vitercik
International Conference on Machine Learning (ICML), 2022
2. **Learning Competitive Equilibria in Exchange Economies with Bandit Feedback**
Wenshuo Guo, Kirthivasan Kandasamy, Joseph E. Gonzalez, Michael Jordan, Ion Stoica
International Conference on Artificial Intelligence and Statistics (AISTATS), 2022.
3. **Learning from an Exploring Demonstrator: Optimal Reward Estimation for Bandits**
Wenshuo Guo, Kumar Krishna Agrawal, Aditya Grover, Vidya Muthukumar, Ashwin Pananjady
International Conference on Artificial Intelligence and Statistics (AISTATS), 2022
4. **Partial Identification with Noisy Covariates: A Robust Optimization Approach**
Wenshuo Guo, Mingzhang Yin, Yixin Wang, Michael I. Jordan
Conference on Causal Learning and Reasoning, 2022.
5. **Robust Learning of Optimal Auctions**
Wenshuo Guo, Michael I. Jordan, Emmanouil Zampetakis
In: *Conference on Neural Information Processing Systems (NeurIPS)* Oral Presentation (top 3% of submissions), 2021.
6. **Test-time Collective Prediction**
Celestine Mendler-Dünnér, **Wenshuo Guo**, Stephen Bates, Michael I. Jordan
In: *Conference on Neural Information Processing Systems (NeurIPS)*, 2021.
7. **The Stereotyping Problem in Collaboratively Filtered Recommender Systems**
Wenshuo Guo, Karl Krauth, Michael I. Jordan, Nikhil Garg.
Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO), 2021
8. **A Variational Inequality Approach to Bayesian Regression Games**
Wenshuo Guo, Michael I. Jordan, Tianyi Lin.
Proceedings of the 60th IEEE Conference on Decision and Control (CDC), 2021

9. **Robust Optimization for Fairness with Noisy Protected Groups**
Serena Wang*, **Wenshuo Guo***, Harikrishna Narasimhan, Andrew Cotter, Maya Gupta, Michael I. Jordan.
In: *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.
10. **Approximate Heavily-Constrained Learning with Lagrange Multiplier Models**
Harikrishna Narasimhan, Andrew Cotter, Yichao Zhou, **Wenshuo Guo**, Serena Wang. 2020. In: *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.
11. **Finding Equilibrium in Multi-Agent Games with Payoff Uncertainty**
Wenshuo Guo, Mihaela Curmei, Serena Wang.
In: *International Conference on Machine Learning (ICML), Workshop on Theoretical Foundations of Reinforcement Learning*, 2020
12. **Fast Algorithms for Computational Optimal Transport and Wasserstein Barycenter**
Wenshuo Guo, Nhat Ho, Michael I. Jordan.
In: *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020, pp. 2088–2097.
13. **Neural Kernel Without Tangents**
Vaishaal Shankar, Alex Fang, **Wenshuo Guo**, Sara Fridovich-Keil, Ludwig Schmidt, Jonathan Ragan-Kelly, Benjamin Recht.
In: *International Conference on Machine Learning (ICML)*, 2020.
14. **Spin Model of Two Random Walkers in Complex Networks**
Wenshuo Guo, Juntao Wang, Kwok Yip Szeto.
In: *International Conference on Complex Networks and Their Applications (CNA)*, 2017.
15. **Optimization of Financial Network Stability by Genetic Algorithm**
Juntao Wang, **Wenshuo Guo**, Kwok Yip Szeto.
In: *IEEE/WIC/ACM International Conference on Web Intelligence (ACM-WI)*, 2017.
16. **Minimization of Systemic Risk for Directed Network Using Genetic Algorithm**
Wenshuo Guo, Kwok Yip Szeto.
In: *European Conference on the Applications of Evolutionary Computation (Evo*)*, 2017.

PREPRINTS

1. **Mechanisms that Incentivize Data Sharing in Federated Learning**
Sai Praneeth Karimireddy*, **Wenshuo Guo***, Michael I. Jordan
Preprint manuscript under review, 2022
2. **Reward Learning as Doubly Nonparametric Bandits: Optimal Design and Scaling Laws** Kush Bhatia, **Wenshuo Guo**, Jacob Steinhardt *Preprint manuscript under review, 2022*

3. **Off-Policy Evaluation with Policy-Dependent Optimization Response**
Wenshuo Guo, Michael I. Jordan, Angela Zhou
Preprint manuscript under review, 2022
4. **Multi-Source Causal Inference Using Control Variates**
Wenshuo Guo*, Serena Wang*, Ding Peng, Yixin Wang, Michael I. Jordan
Preprint manuscript under review, 2022
5. **Cilantro: Online Learning for Performance-Aware Resource Allocation**
Romil Bhardwaj, Kirthivasan Kandasamy, Wenshuo Guo, Asim Biswal, Joseph E. Gonzalez, Benjamin Hindman, Ion Stoica
Preprint manuscript under review, 2022
6. **Evaluating Recommender Systems under Distribution Shifts and Feedback Loops**
Karl Krauth, Sarah Dean*, Alex Zhao*, Wenshuo Guo*, Mihaela Curmei*, Benjamin Recht, Michael I. Jordan.
Under review at Journal of Machine Learning Research (JMLR), 2022.

INVITED TALKS

- **Towards Credible Data-Driven Decision-Making under Social and Economic Needs (50 mins virtual talk)**
at Google Brain, Mountain View, 2022.
- **No-Regret Learning in Partially-Informed Auctions**
at NYU Stern Operations Management group, 2022.
- **Machine Learning meets Economics: Learning under Social and Economic Needs (50 mins virtual talk)**
at Morgan Stanley Machine Learning Research, New York, 2022.
- **Towards Adaptive and Robust Learning in Data-Driven Mechanism Design (50 mins spotlight)**
at UC Berkeley BAIR/CPAR/BDD seminar, hosted by the Berkeley Artificial Intelligence Research Lab, in conjunction with CITRIS People & Robots and Berkeley DeepDrive, 2021.
- **Robust Learning of Optimal Auctions**
at Google PhD Fellowship Summit, 2021.
- **Learning from an Exploring Demonstrator: Optimal Reward Estimation for Bandits**
at ICML Workshop on Human-AI Collaboration in Sequential Decision-Making (spotlight talk), 2021.

SKILLS

- Programming: Python, C++, SQL, PyTorch, TensorFlow, Git, CSS, Mathematica, Matlab.
- Language: English (native / bilingual proficiency), Chinese (native / bilingual proficiency), Cantonese (limited working proficiency).